Progress Report

Discrete State-Space Models and Expectation-Maximization for Cell-State Trajectories

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Overview

The goal of the project is to model how macrophages reprogram into mural-like cells during tissue healing, using a Hidden Markov Model (HMM) fitted with the EM algorithm (Baum–Welch) on single-cell RNA-seq data.

During the first two weeks I followed the reading guide and focused on the mathematical foundations. Progress against the suggested plan:

- Probability basics, Markov chains, likelihood and MLE: **done**.
- Hidden Markov Model (Rabiner, Sections II–III): **done**. The three basic problems still need more work.
- Hidden variables and the EM idea: **started**.
- Bilmes (1998) and toy implementation of forward–backward / Baum–Welch: **not yet**.

According to the reading guide I am about one week behind. The rest of this report summarises what I understand so far and how each part connects to the project.

1 Probability basics and Bayes' theorem

What I learned

Conditional probability $P(A \mid B)$ is the probability of A once B is known. Two events are independent if knowing one tells nothing about the other. Bayes' theorem turns $P(B \mid A)$ into $P(A \mid B)$:

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}.$$

In the project

The model gives $P(\text{gene expression} \mid \text{cell state})$. The question I want to answer is the reverse, $P(\text{cell state} \mid \text{gene expression})$:

$$P(\text{cell state} \mid \text{gene expression}) = \frac{P(\text{gene expression} \mid \text{cell state})\,P(\text{cell state})}{P(\text{gene expression})}.$$

Bayes' theorem makes this reversal possible, and the forward–backward algorithm is Bayes' theorem applied over a whole sequence.

2 Markov chains and the transition matrix

What I learned

The Markov property: the next state depends only on the current state, not on the whole history,

$$P(s_{t+1} \mid s_t, s_{t-1}, \ldots, s_1) = P(s_{t+1} \mid s_t).$$

A Markov chain is a sequence of states following this rule. The transition matrix A has one row per current state and one column per next state, $A[i,j] = P(s_{t+1} = j \mid s_t = i)$, and each row sums to 1. Multiplying a state distribution by A gives the distribution one step later.

In the project

The cell states (Macrophage, Transitional, Mural-like) form the Markov chain. $A[i,j]$ is the probability that a cell in state i is in state j at the next time point. The Markov assumption means a cell's next step depends only on its current state, not on how it got there. The learned A is the main result of the project: it shows the reprogramming path, e.g. Macrophage $\rightarrow$ Transitional $\rightarrow$ Mural-like, and how fast it happens.

3 Likelihood and maximum likelihood

What I learned

The likelihood $L(\theta) = P(\text{data} \mid \theta)$ is the probability of the observed data under a chosen parameter value θ, read as a function of θ while the data stays fixed. Maximum likelihood estimation picks the θ that gives the observed data the highest probability. It is the same formula as the probability, only read the other way around.

In the project

The parameters are $\theta = (\pi, A, B)$, where B contains the Gaussian emission parameters (mean and variance of gene expression per state). The likelihood $P(O \mid \theta)$ says how well these parameters explain the observed cell data. Baum–Welch is an algorithm that increases this likelihood step by step.

4 Hidden variables and the EM idea

What I learned

A hidden variable is a variable that affects the data but is never observed. With hidden variables, maximum likelihood cannot be solved directly. EM handles this with two repeated steps:

- **E-step:** using the current parameters, estimate the hidden variables as probabilities.
- **M-step:** using these estimates, re-estimate the parameters as if the hidden variables were known.

Each round increases (or keeps) the likelihood.

In the project

The hidden variable is the cell state of each cell; only the gene expression is observed. The E-step estimates, for each cell, the probability of being in each state ($\gamma_t(i)$) and the probability of each transition ($\xi_t(i,j)$). The M-step turns these into new values for π, A and the Gaussians.

5 The Hidden Markov Model (Rabiner, Sections II–III)

What I learned

An HMM is a Markov chain over hidden states $s_1 \dots s_T$, plus an observation o_t at each time step that depends only on the current state. It is defined by $\lambda = (\pi, A, B)$. The probability of one full path is

$$P(s_1 \dots s_T,\ o_1 \dots o_T \mid \lambda) = \pi(s_1)\, B[s_1, o_1] \prod_{t=2}^{T} A[s_{t-1}, s_t]\, B[s_t, o_t].$$

Rabiner's three basic problems:

1. **Evaluation:** compute $P(O \mid \lambda)$ by summing over all state paths. The forward algorithm does this efficiently with a table $\alpha_t(i)$ over states and time, instead of listing every path.
2. **Decoding:** find the single most likely state path (Viterbi). Same table idea, but taking the maximum instead of the sum.
3. **Learning:** find π, A, B from observations alone (Baum–Welch). This is EM for HMMs: the forward table α and backward table β together give γ and ξ (E-step), which re-estimate the parameters (M-step).

In the project

The mapping between the HMM and the project:

HMM element	In Rabiner	In the project
Hidden state s_t	State of the system at time t (not observed)	Cell state: Macrophage, Transitional, or Mural-like
N	Number of states	$N = 3$
Observation o_t	What is measured at time t	Gene-expression profile of a cell (a vector of expression values)
π	Probability of the first state	Which state a cell starts in (mostly Macrophage after injury)
A (transition matrix)	$P(s_{t+1} = j \mid s_t = i)$	How cells reprogram, e.g. P(Macrophage $\to$ Transitional). The main result of the project
B (emission model)	$P(o_t \mid s_t)$	A Gaussian over gene expression for each state, since expression values are continuous
Observation sequence O	$o_1 \dots o_T$ in time order	Ordered cell observations across time points (how to build this ordering is my open question)

Problem 3 is the core of the project: learning A from the data. Problem 1 gives the likelihood used to check that training is improving. Problem 2 could be used later to label each cell with its most likely state. I understand what the three problems are and how they relate, but I still have to go through their solutions in Rabiner in detail.

6 What is still unclear to me

- The solutions to Rabiner's three problems in full detail (forward, Viterbi, Baum–Welch), and the exact derivation of the Baum–Welch re-estimation formulas, and the Gaussian-emission version of the M-step (Bilmes). I understand the idea but have not worked through the equations in full yet.
- How to build the observation sequence O. scRNA-seq measures each cell only once, so no single cell is followed over time. I am unsure whether the sequence should come from the time points in the dataset, from a pseudotime ordering, or something else.
- Numerical issues in forward–backward (probabilities becoming very small over long sequences) and how they are usually handled.

7 Plan for the next two weeks

- Finish Bilmes (1998): the general EM recipe and the HMM/Gaussian update equations.
- Implement forward–backward and Baum–Welch on a small toy sequence, as suggested in the reading guide.
- Discuss the sequence definition with the supervisor before touching the real data.